

Design of Hybrid Human–AI Agent Organizations: A Mathematical Framework for Organizational Dynamics

Usman Zafar, Ph.D
Founder, zulfr.com
Contact: info@zulfr.com

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Abstract

This work develops a mathematically closed, fully identifiable, and empirically falsifiable theory of hybrid Human–AI organizational design. The research introduces a unified probabilistic representation in which classical and AI-native organizational dimensions are embedded into a 17-dimensional latent space and mapped to a Fisher-geometric probability simplex through a softmax transformation. Measurement is formalized through RKHS-bounded feature maps, dynamics evolve on latent space via controlled stochastic differential equations, and governance is encoded through a Kullback–Leibler potential with strict dissipation. The resulting system forms a controlled stochastic manifold with explicit information geometry and guarantees convergence to a governance-optimal equilibrium.

Note: Definitions are provided in the Appendices.

Keywords: Organization Theory and Design; Hybrid Human–AI Organizations; Latent Probabilistic Representation; Softmax Mapping; Probability Simplex; Fisher Information Geometry; Kullback–Leibler Divergence; Governance Potential; Controlled Stochastic Differential Equations (SDEs);

RKHS Feature Maps; Information-Theoretic Normalization; Mixed-Data Likelihood Models; Stochastic Manifold; Governance-Optimal Equilibrium; Exponential Convergence; Organizational Measurement Closure.

1 Introduction

Hybrid Human–AI organizations are inherently complex systems, combining human decision-making, AI autonomy, structural dependencies, and governance requirements. Classical organizational theory provides useful dimensions but lacks a mathematically closed framework capable of capturing AI-native behavior, uncertainty, and information theoretic structure. This research introduces a new approach: instead of increasing structural complexity, it reduces the organizational system to a simple and unified probabilistic representation. The development of this framework involved evaluating two distinct normalization strategies.

The first relied on unit-interval normalization, where each organizational dimension was scaled to the range $(0, 1)$. Although Simple, this approach proved inadequate for large and heterogeneous organizational setups, as it clipped information, could not represent uncertainty, and was incompatible with information-geometric tools. The second and final approach introduced a latent probabilistic representation, in which each organizational dimension is generated by a latent variable and mapped to the probability simplex through a softmax transformation. This representation generalizes across diverse organizational contexts, supports mixed data types, and provides a natural foundation for Fisher–KL geometry and stochastic governance dynamics. It also ensures automatic simplex invariance and enables uncertainty-aware modeling.

The resulting framework unifies measurement, dynamics, constraints, and governance into a single information-geometric system. Measurement is closed through RKHS-bounded feature maps, dynamics evolve on latent space via controlled stochastic differential equations, and governance is encoded through a Kullback–Leibler potential with strict dissipation. Together, these elements form a controlled stochastic manifold with explicit Fisher–KL information geometry and guarantee convergence to a governance-optimal equilibrium. Despite the depth of the underlying mathematics, the final structure is simple, interpretable, and empirically testable. It provides the first mathematically indestructible foundation for AI-native organizational

theory and design.

2 Preliminaries and Notation

Let $\mathcal{U}(t)$ denote the organizational universe at time t , including documents, communications, logs, code, and operational traces. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_t\}$ generated by $\mathcal{U}(s)$ for $s \leq t$.

We denote by $x(t) \in [0, 1]^{17}$ the hybrid organizational state, consisting of classical structural dimensions, classical contextual dimensions, and AI-native dimensions. All coordinates are normalized to $[0, 1]$ and embedded into a Fisher-weighted simplex.

We use $D_{\text{KL}}(p||q)$ for Kullback–Leibler divergence, $\mathcal{I}(x)$ for Fisher information, and $\mathcal{L}$ for the infinitesimal generator of a controlled SDE.

3 Hybrid Human–AI Organizational State Space

3.1 State vector and dimensions

We represent the organization at time t by a 17-dimensional state vector

$$x(t) \in \mathbb{R}^{17}.$$

The classical structural dimensions are

$$x^{\text{str}}(t) = (F(t), S_p(t), H(t), C(t), P(t), R(t)) \in [0, 1]^6,$$

where F (formalization), S_p (specialization), H (hierarchy), C (centralization), P (professionalism), R (personnel ratio index) are all normalized to $[0, 1]$.

The classical contextual dimensions are

$$x^{\text{ctx}}(t) = (G_s(t), E(t), Z(t), U(t), T(t)) \in [0, 1]^5,$$

where G_s (goals/strategy), E (environment), Z (size), U (culture), T (technology) are normalized to $[0, 1]$.

The AI-native dimensions are

$$x^{\text{AI}}(t) = (AP(t), AA(t), ADS(t), ACD(t), CLR(t), AGC(t)) \in [0, 1]^6,$$

where AP (AI penetration), AA (AI autonomy), ADS (AI decision share), ACD (AI coordination density), CLR (cognitive load redistribution), AGC (AI governance constraint) are normalized to $[0, 1]$.

The full hybrid state is

$$x(t) = (x^{\text{str}}(t), x^{\text{ctx}}(t), x^{\text{AI}}(t)) \in [0, 1]^{17}.$$

3.2 Latent Probabilistic State Representation

The classical normalization $x_i(t) \in [0, 1]$ is replaced by a mixed-data latent probabilistic representation consistent with information geometry.

Latent variables. Each organizational dimension is generated by a latent variable

$$z_i(t) = \beta_i^\top \phi(U(t)) + \epsilon_i(t), \quad i = 1, \dots, 17,$$

where $\phi(U(t))$ is an embedding of the organizational universe (documents, logs, communications, metrics), and $\epsilon_i(t)$ is noise.

Mixed-data likelihood. The observation model for each dimension is chosen according to data type:

$$p(x_i^{\text{obs}} | z_i) = \begin{cases} \mathcal{N}(x_i | z_i, \sigma_i^2), & \text{quantitative, 4pt] ordered logit/probit,} \\ \text{ordinal, 4pt] categorical softmax,} & \text{nominal, 4pt] text embedding likelihood,} \\ \text{text.} & \end{cases}$$

The unified likelihood is

$$\mathcal{L} = \prod_{i=1}^{17} p(x_i^{\text{obs}} | z_i(t)).$$

Uncertainty-aware latent state. Each latent coordinate is modeled as

$$z_i(t) \sim \mathcal{N}(\mu_i(t), \sigma_i^2(t)).$$

Mapping to the simplex. The organizational state is defined as the softmax-normalized latent vector:

$$p_i(t) = \frac{\exp(z_i(t))}{\sum_{j=1}^{17} \exp(z_j(t))}, \quad p(t) \in \Delta_{16}.$$

Final upgraded state.

$$x(t) \equiv p(t) = \text{softmax}(z(t)), \quad z(t) \sim \mathcal{N}(\mu(t), \Sigma(t)).$$

This representation preserves information, handles mixed qualitative and quantitative data, supports uncertainty, and is fully compatible with KL divergence, Fisher information geometry, and the stochastic manifold structure of the organizational dynamics.

3.3 Measurement closure

Let $\mathcal{U}(t)$ denote the organizational universe at time t (documents, communications, logs, code, metrics, etc.). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_t\}_{t \geq 0}$ a filtration representing the information available up to time t , generated by $\mathcal{U}(s)$ for $s \leq t$.

For each dimension $i = 1, \dots, 17$, define a (possibly nonlinear) feature map

$$\varphi_i : \mathcal{U} \rightarrow \mathbb{R},$$

implemented in practice by LLM-based semantic extraction.

We define the *observable* organizational state as the conditional expectation

$$x_i(t) = \mathbb{E}[\varphi_i(\mathcal{U}) \mid \mathcal{F}_t], \quad i = 1, \dots, 17.$$

Thus, each dimension $x_i(t)$ is a well-defined $\mathcal{F}_t$ -adapted process and is, in principle, statistically estimable from data, rather than arbitrarily assigned. The LLM layer is operationally embedded in the choice of φ_i , but the state itself is defined in purely probabilistic terms.

3.4 State constraint: compositional structure

To remove scale ambiguity and enforce trade-offs among dimensions, we impose a compositional (simplex) constraint on the state:

$$\sum_{i=1}^{17} w_i x_i(t) = 1, \quad x_i(t) \in [0, 1], \quad w_i > 0, \quad \sum_{i=1}^{17} w_i = 1.$$

Here w_i are fixed, exogenously chosen importance weights (e.g., reflecting strategic priorities or governance preferences). This constraint makes the system identifiable up to a known, explicit weighting scheme and forces any increase in some dimensions to be compensated by decreases (or slower growth) in others.

Define the admissible state space

$$\mathcal{X} = \left\{ x \in [0, 1]^{17} : \sum_{i=1}^{17} w_i x_i = 1 \right\}.$$

3.5 Dynamic model and falsifiability

We model the evolution of the organizational state as a controlled stochastic differential equation (SDE) on $\mathcal{X}$:

$$dx(t) = f(x(t), u(t); \theta) dt + \Sigma(x(t); \theta) dW_t,$$

where

- $x(t) \in \mathcal{X}$ is the state,
- $u(t)$ is a control/design input (e.g., structural changes, AI deployment, governance actions),
- W_t is a k -dimensional standard Brownian motion,
- $f : \mathcal{X} \times \mathcal{U}_u \rightarrow \mathbb{R}^{17}$ is the drift,
- $\Sigma : \mathcal{X} \rightarrow \mathbb{R}^{17 \times k}$ is the diffusion,
- θ is a finite-dimensional parameter vector.

We impose the following conditions:

1. **Lipschitz and growth conditions.** The functions $f(\cdot, \cdot; \theta)$ and $\Sigma(\cdot; \theta)$ are globally Lipschitz in x and satisfy linear growth bounds, ensuring existence and uniqueness of strong solutions for given (u, θ) .
2. **State invariance.** The dynamics preserve the simplex constraint:

$$x(0) \in \mathcal{X} \quad \Rightarrow \quad x(t) \in \mathcal{X} \quad \text{for all } t \geq 0,$$

which can be enforced by constructing f and Σ tangent to $\mathcal{X}$.

3. **Parameter identifiability.** The parameter vector θ is identifiable from observed trajectories $\{x(t)\}_{t \in [0, T]}$ under suitable excitation of $u(t)$, i.e., distinct parameter values yield distinguishable distributions over paths.
4. **Predictive falsifiability.** For a given estimated parameter $\hat{\theta}$ and control $u(t)$, the model yields a predictive distribution for $x(t + \Delta)$:

$$\hat{x}(t + \Delta) = \mathbb{E}_{\hat{\theta}}[x(t + \Delta) \mid \mathcal{F}_t].$$

The model is empirically testable via the requirement

$$\hat{x}(t + \Delta) \approx x(t + \Delta)$$

in an appropriate statistical sense (e.g., mean squared prediction error, likelihood-based tests). Systematic, significant deviations provide grounds to reject or revise the model.

3.6 Dynamic model and falsifiability (Latent-Probabilistic Form)

The organizational state is represented by latent variables

$$z(t) \in \mathbb{R}^{17}, \quad x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16}.$$

The dynamics evolve on the unconstrained latent space:

$$dz(t) = f_z(z(t), u(t); \theta) dt + \Sigma_z(z(t); \theta) dW_t,$$

where

- $z(t)$ is the latent organizational state,

- $x(t) = \text{softmax}(z(t))$ is the induced probability state,
- $u(t) \in \mathcal{U}_{\text{adm}}$ is a governance/structural control input,
- W_t is a k -dimensional Brownian motion,
- f_z and Σ_z are drift and diffusion in latent space,
- θ is a finite-dimensional parameter vector.

Softmax mapping and induced dynamics. The observable state satisfies

$$x_i(t) = \frac{\exp(z_i(t))}{\sum_{j=1}^{17} \exp(z_j(t))}.$$

By Itô's lemma, the induced dynamics on $x(t)$ are

$$dx(t) = J(z(t)) dz(t) + \frac{1}{2} H(z(t)) : \Sigma_z(z(t)) \Sigma_z(z(t))^\top dt,$$

where J and H are the Jacobian and Hessian of the softmax map.

No simplex constraint needed. Because $x(t) = \text{softmax}(z(t))$,

$$x(t) \in \Delta_{16} \quad \text{for all } t,$$

and the tangent constraints of the earlier formulation are no longer required.

Parameter identifiability. The parameter vector θ is identifiable from latent trajectories

$$\{z(t)\}_{t \in [0, T]}$$

or equivalently from observed trajectories $\{x(t)\}$ via the softmax likelihood.

Predictive falsifiability. Given an estimated parameter $\hat{\theta}$,

$$\hat{z}(t + \Delta) = \mathbb{E}_{\hat{\theta}}[z(t + \Delta) \mid \mathcal{F}_t], \quad \hat{x}(t + \Delta) = \text{softmax}(\hat{z}(t + \Delta)).$$

The model is falsifiable if

$$\hat{x}(t + \Delta) \not\approx x(t + \Delta)$$

under likelihood or divergence-based tests.

3.7 Core (0,1) Unit Interval Normalization

Under the measurement closure (conditional expectation with respect to $\mathcal{F}_t$), the compositional state constraint (simplex structure on $\mathcal{X}$), and the controlled SDE with identifiable parameters and explicit predictive tests, the hybrid human–AI organizational system is:

- *measurable*: each dimension $x_i(t)$ is a well-defined stochastic process adapted to $\mathcal{F}_t$;
- *constrained*: the state evolves within a known, compact, compositional manifold $\mathcal{X}$;
- *falsifiable*: the dynamics generate testable predictions over trajectories.

Anything less is purely descriptive; this construction yields a scientifically testable 17-dimensional organizational design system for AI-native enterprises.

3.8 Core (Latent–Probabilistic Version)

Under the measurement closure via RKHS feature maps, the latent–probabilistic state representation

$$z(t) \sim \mathcal{N}(\mu(t), \Sigma(t)), \quad x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16},$$

and the controlled SDE on latent space

$$dz(t) = f_z(z(t), u(t); \theta) dt + \Sigma_z(z(t); \theta) dW_t,$$

the hybrid human–AI organizational system satisfies the following properties:

Measurable. Each latent coordinate $z_i(t)$ is an $\mathcal{F}_t$ -adapted stochastic process, and the observable state $x(t) = \text{softmax}(z(t))$ is a measurable transformation of $z(t)$.

Constrained. Because $x(t) = \text{softmax}(z(t))$, the state always lies in the probability simplex:

$$x(t) \in \Delta_{16} \quad \text{for all } t,$$

without requiring explicit tangent constraints.

Uncertainty-aware. The latent distribution $z(t) \sim \mathcal{N}(\mu(t), \Sigma(t))$ provides a full probabilistic description of organizational uncertainty, rather than a clipped $[0, 1]$ scalar.

Information-geometric. The Fisher information of $p(t)$ induces the natural Riemannian metric, and the KL governance potential

$$V(x) = D_{\text{KL}}(p(x) \parallel p(x^*))$$

is the canonical Lyapunov function on this manifold.

Falsifiable. Given an estimated parameter $\hat{\theta}$,

$$\hat{z}(t + \Delta) = \mathbb{E}_{\hat{\theta}}[z(t + \Delta) \mid \mathcal{F}_t], \quad \hat{x}(t + \Delta) = \text{softmax}(\hat{z}(t + \Delta)),$$

and the model is falsifiable whenever

$$\hat{x}(t + \Delta) \not\approx x(t + \Delta)$$

under likelihood or divergence-based tests.

Scientifically testable. The system forms a controlled stochastic manifold with:

- identifiable parameters,
- measurable latent processes,

- automatic simplex invariance,
- KL-based governance contraction,
- and explicit predictive tests.

Anything less is descriptive; this construction yields a fully probabilistic, information-geometric, and empirically falsifiable organizational theory for AI-native enterprises.

3.9 Information-theoretic closure and invariants(Unit Interval Normalization)

The previous construction yields a measurable, constrained, and falsifiable 17-dimensional system, but three degrees of arbitrariness remain: (i) exogenous weights w_i , (ii) unconstrained feature maps φ_i , (iii) no structural invariant beyond the simplex constraint. We now remove these.

3.9.1 Endogenous information-theoretic weights

Let $x_i(t)$ be the i -th coordinate of the state and let θ denote the parameter vector of the dynamics. Consider the Fisher information of x_i with respect to θ over a time window $[0, T]$:

$$\mathcal{I}_i(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p(x_i(0 : T) \mid \theta) \right)^2 \right], \quad i = 1, \dots, 17,$$

where $p(x_i(0 : T) \mid \theta)$ is the marginal likelihood of the trajectory segment $x_i(0 : T)$.

Define the information weight

$$\lambda_i = \mathcal{I}_i(\theta), \quad \lambda_i \geq 0,$$

and normalize to obtain endogenous compositional weights

$$w_i = \frac{\lambda_i}{\sum_{j=1}^{17} \lambda_j}, \quad \sum_{i=1}^{17} w_i = 1, \quad w_i \geq 0.$$

The simplex constraint becomes

$$\sum_{i=1}^{17} w_i x_i(t) = 1, \quad x_i(t) \in [0, 1],$$

with w_i now determined by the information content of each dimension rather than arbitrarily chosen. Dimensions that are more informative about the system parameters (and thus about the dynamics) receive higher weight.

3.10 Information-Theoretic Closure and Invariants (Latent Probabilistic Version)

The latent-probabilistic state representation

$$z(t) \sim \mathcal{N}(\mu(t), \Sigma(t)), \quad x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16},$$

removes the remaining degrees of arbitrariness in the earlier formulation. We now close the system information-theoretically.

3.10.1 Endogenous Fisher-information weights

Because the observable state is $p(t) = \text{softmax}(z(t))$, the natural information measure is the Fisher information of the induced probability distribution:

$$\mathcal{I}(p) = \mathbb{E}[\nabla_z \log p(x^{\text{obs}} | z) \nabla_z \log p(x^{\text{obs}} | z)^\top].$$

Define the coordinate-wise Fisher information:

$$\lambda_i = \mathcal{I}_{ii}(p), \quad \lambda_i \geq 0.$$

Normalize to obtain endogenous weights:

$$w_i = \frac{\lambda_i}{\sum_{j=1}^{17} \lambda_j}, \quad \sum_{i=1}^{17} w_i = 1.$$

These weights are no longer arbitrary: they reflect the sensitivity of the likelihood to changes in the latent parameters and therefore encode the true information contribution of each organizational dimension.

3.10.2 Mixed-data likelihood and feature constraints

Each observed dimension is generated from the latent variable $z_i(t)$ via a mixed-data likelihood:

$$p(x_i^{\text{obs}} | z_i) = \begin{cases} \mathcal{N}(x_i | z_i, \sigma_i^2), & \text{quantitative,} \\ \text{Theunifiedlikelihood is } \mathcal{L} = \prod_{i=1}^{17} p(x_i^{\text{obs}} | z_i(t)). \end{cases}$$

Feature maps φ_i used in measurement closure satisfy

$$\varphi_i \in \mathcal{H}, \quad \|\varphi_i\|_{\mathcal{H}} \leq C,$$

ensuring compactness and eliminating representational arbitrariness.

3.10.3 Information-geometric invariants

Because $x(t) = \text{softmax}(z(t))$ lies in the exponential family, the Fisher information induces the natural Riemannian metric on the simplex. The KL divergence

$$V(x) = D_{\text{KL}}(p(x) \| p(x^*))$$

is the canonical potential associated with this geometry.

At equilibrium:

$$\nabla^2 V(x^*) = \mathcal{I}(x^*),$$

establishing the Fisher–KL duality and giving the manifold its intrinsic curvature.

3.10.4 Final structure

The hybrid human–AI organization is now characterized by:

- a latent Gaussian state $z(t)$ with uncertainty,
- a softmax-mapped probability state $x(t) \in \Delta_{16}$,
- endogenous Fisher-information weights w_i ,
- RKHS-bounded feature maps for measurement closure,

- a mixed-data likelihood consistent with all variable types,
- and a KL governance potential aligned with the Fisher geometry.

This removes all remaining arbitrariness and yields a fully information-geometric, probabilistic, and governance-aligned organizational state representation.

4 Feature maps in a constrained RKHS Unit Interval Normalization

Let $\mathcal{H}$ be a reproducing kernel Hilbert space (RKHS) over the organizational universe $\mathcal{U}$, with kernel $k(\cdot, \cdot)$ and norm $\|\cdot\|_{\mathcal{H}}$. We constrain each feature map φ_i to lie in $\mathcal{H}$ with bounded norm:

$$\varphi_i \in \mathcal{H}, \quad \|\varphi_i\|_{\mathcal{H}} \leq C, \quad i = 1, \dots, 17,$$

for some fixed constant $C > 0$.

The observable state remains

$$x_i(t) = \mathbb{E}[\varphi_i(\mathcal{U}) \mid \mathcal{F}_t],$$

but the hypothesis space for φ_i is now a compact, regularized function class. This removes hidden degrees of freedom in the representation and enables standard tools from statistical learning theory (e.g., generalization bounds, capacity control).

4.0.1 Energy/entropy invariant and governance stability

Let $V : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ be a continuously differentiable functional (Lyapunov-like), interpreted as an energy or entropy of the organizational state (e.g., misalignment, risk, or governance tension):

$$V = V(x).$$

We impose a *governance stability law*:

$$\frac{d}{dt} \mathbb{E}[V(x(t))] \leq 0,$$

for all admissible controls $u(t)$ that satisfy governance constraints. Equivalently, in infinitesimal generator form for the SDE

$$dx(t) = f(x(t), u(t); \theta) dt + \Sigma(x(t); \theta) dW_t,$$

with generator

$$\mathcal{L}V(x) = \nabla V(x)^\top f(x, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma(x; \theta) \Sigma(x; \theta)^\top \nabla^2 V(x)),$$

we require

$$\mathcal{L}V(x) \leq 0 \quad \text{for all } x \in \mathcal{X} \text{ and admissible } u.$$

This condition enforces that, in expectation, the system does not increase its governance energy/entropy over time; it either remains stable or dissipates toward a governed manifold (e.g., a set of acceptable configurations).

4.0.2 Resulting structure: controlled stochastic manifold with information geometry

The hybrid human–AI organization is now characterized by:

- a constrained state space $\mathcal{X} \subset [0, 1]^{17}$ with an endogenous, information-theoretic simplex structure;
- observable coordinates $x_i(t)$ defined as conditional expectations of RKHS-bounded feature maps φ_i ;
- controlled stochastic dynamics

$$dx(t) = f(x(t), u(t); \theta) dt + \Sigma(x(t); \theta) dW_t,$$

with identifiable parameters and Lyapunov-type governance invariant $V(x)$ satisfying $\frac{d}{dt} \mathbb{E}[V(x(t))] \leq 0$.

This yields a controlled stochastic manifold endowed with an information-geometric structure (via Fisher information weights) and a governance-stability law. At this point, the construction crosses from a descriptive framework into a falsifiable, mathematically grounded theory of AI-native organizational design.

5 Feature Maps + Invariants (Latent–Probabilistic Version)

Let $\mathcal{H}$ be a reproducing kernel Hilbert space (RKHS) over the organizational universe $\mathcal{U}$, with kernel $k(\cdot, \cdot)$ and norm $\|\cdot\|_{\mathcal{H}}$. Each feature map

$$\varphi_i : \mathcal{U} \rightarrow \mathbb{R}$$

is constrained to lie in $\mathcal{H}$ with bounded norm:

$$\varphi_i \in \mathcal{H}, \quad \|\varphi_i\|_{\mathcal{H}} \leq C.$$

The latent state is defined by

$$z_i(t) = \beta_i^\top \varphi(U(t)) + \epsilon_i(t), \quad i = 1, \dots, 17,$$

and the observable organizational state is the softmax-normalized vector

$$x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16}.$$

The RKHS constraint ensures compactness of the hypothesis space and eliminates representational arbitrariness while supporting mixed-data embeddings (quantitative, ordinal, nominal, text).

5.1 Governance stability via KL potential

Let x^* denote the governance-optimal equilibrium and define the governance potential

$$V(x) = D_{\text{KL}}(p(x) \| p(x^*)).$$

Because $x(t) = \text{softmax}(z(t))$, the dynamics evolve on latent space:

$$dz(t) = f_z(z(t), u(t); \theta) dt + \Sigma_z(z(t); \theta) dW_t.$$

By Itô's lemma, the induced generator acting on $V(x)$ is

$$\mathcal{L}V(x) = \nabla_z V(x)^\top f_z(z, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma_z(z; \theta) \Sigma_z(z; \theta)^\top \nabla_z^2 V(x)).$$

We impose the governance stability law:

$$\mathcal{L}V(x) \leq 0 \quad \text{for all } x \in \Delta_{16}, u \in \mathcal{U}_{\text{adm}}.$$

This ensures that the expected KL divergence to the governance-optimal state does not increase over time.

5.2 Resulting structure: controlled stochastic manifold with information geometry

The hybrid human–AI organization is now characterized by:

- a latent Gaussian state $z(t)$ with uncertainty,
- a softmax-mapped probability state $x(t) \in \Delta_{16}$,
- RKHS-bounded feature maps φ_i ensuring compactness,
- endogenous Fisher-information weights defining the simplex geometry,
- a KL governance potential aligned with the Fisher metric,
- controlled stochastic dynamics on latent space,
- and a generator $\mathcal{L}$ guaranteeing governance dissipation.

This yields a controlled stochastic manifold endowed with natural information geometry (Fisher–KL duality) and a governance-stability law. The construction is fully probabilistic, non-arbitrary, and empirically falsifiable, completing the transition from descriptive framework to mathematically grounded theory.

6 Governance potential as KL divergence Unit Interval Normalization

In the previous subsection, the existence of a Lyapunov-like functional V was assumed. We now make V explicit and governance-grounded, removing the remaining arbitrariness.

Let $\mathcal{X} \subset [0, 1]^{17}$ be the simplex-constrained state space with endogenous information-theoretic weights w_i . Assume that governance specifies a unique *optimal equilibrium state*

$$x^* \in \mathcal{X},$$

interpreted as the target (governance-optimal) configuration of the 17 dimensions.

We treat x and x^* as probability vectors on the index set $\{1, \dots, 17\}$, e.g. by

$$p_i(x) = w_i x_i, \quad p_i(x^*) = w_i x_i^*, \quad \sum_{i=1}^{17} p_i(x) = \sum_{i=1}^{17} p_i(x^*) = 1.$$

Define the governance potential $V : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ as the Kullback–Leibler divergence between the current weighted state and the governance-optimal state:

$$V(x) = D_{\text{KL}}(p(x) \parallel p(x^*)) = \sum_{i=1}^{17} p_i(x) \log \frac{p_i(x)}{p_i(x^*)}.$$

By construction, $V(x) \geq 0$ and $V(x) = 0$ if and only if $p(x) = p(x^*)$, i.e. $x = x^*$ (up to degenerate weights).

6.0.1 Generator condition and convergence

Recall the SDE

$$dx(t) = f(x(t), u(t); \theta) dt + \Sigma(x(t); \theta) dW_t,$$

with infinitesimal generator

$$\mathcal{L}V(x) = \nabla V(x)^\top f(x, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma(x; \theta) \Sigma(x; \theta)^\top \nabla^2 V(x)).$$

We impose the *governance stability law* in its explicit KL form:

$$\mathcal{L}V(x) \leq 0 \quad \text{for all } x \in \mathcal{X} \text{ and admissible } u.$$

Under standard conditions (e.g., positive recurrence, compactness of $\mathcal{X}$, and $\mathcal{L}V(x) < 0$ for $x \neq x^*$), this yields:

$$x(t) \rightarrow x^* \quad \text{in probability (or almost surely), as } t \rightarrow \infty.$$

Thus:

- the Lyapunov function V is no longer arbitrary; it is uniquely tied to the governance-optimal equilibrium via D_{KL} ;
- governance is encoded as a target distribution $p(x^*)$ over dimensions, not just as constraints;
- stability, convergence, and interpretability are unified: the organization is driven (in expectation) to reduce its divergence from the governance-optimal state.

6.0.2 Final structure

The resulting system is:

- a 17-dimensional, simplex-constrained, information-weighted state space $\mathcal{X}$;
- with observable coordinates $x_i(t)$ defined as conditional expectations of RKHS-bounded feature maps;
- evolving under a controlled SDE with identifiable parameters;
- endowed with an information-geometric governance potential

$$V(x) = D_{\text{KL}}(p(x) \parallel p(x^*)),$$

satisfying $\mathcal{L}V(x) \leq 0$ and implying $x(t) \rightarrow x^*$.

This closes the remaining degrees of freedom: measurement, constraints, dynamics, and governance are now jointly specified in a controlled stochastic manifold with explicit information geometry.

6.1 Governance Potential as KL Divergence (Latent–Probabilistic Version)

In the latent–probabilistic formulation, the organizational state is

$$z(t) \sim \mathcal{N}(\mu(t), \Sigma(t)), \quad x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16}.$$

Let x^* denote the governance-optimal equilibrium, defined as the minimizer of a governance cost functional:

$$x^* = \arg \min_{x \in \Delta_{16}} \mathbb{E}[C(x)].$$

Assume strict interiority:

$$p_i(x^*) \geq \varepsilon > 0, \quad \forall i,$$

ensuring that KL divergence is finite and differentiable.

6.2 KL governance potential

Define the governance potential

$$V(x) = D_{\text{KL}}(p(x) \parallel p(x^*)) = \sum_{i=1}^{17} p_i(x) \log \frac{p_i(x)}{p_i(x^*)}.$$

This potential satisfies:

$$V(x) \geq 0, \quad V(x) = 0 \iff x = x^*.$$

At equilibrium, Fisher–KL duality holds:

$$\nabla^2 V(x^*) = \mathcal{I}(x^*),$$

where $\mathcal{I}$ is the Fisher information matrix of $p(x)$.

6.3 Generator condition and convergence

The dynamics evolve on latent space:

$$dz(t) = f_z(z(t), u(t); \theta) dt + \Sigma_z(z(t); \theta) dW_t.$$

By Itô's lemma, the generator acting on $V(x)$ is

$$\mathcal{L}V(x) = \nabla_z V(x)^\top f_z(z, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma_z(z; \theta) \Sigma_z(z; \theta)^\top \nabla_z^2 V(x)).$$

We impose the governance stability law:

$$\mathcal{L}V(x) \leq 0 \quad \forall x \in \Delta_{16}, \quad u \in \mathcal{U}_{\text{adm}}.$$

Under strict negativity,

$$\mathcal{L}V(x) < 0 \quad \forall x \neq x^*,$$

the system satisfies

$$x(t) \rightarrow x^* \quad \text{almost surely and in probability.}$$

Under the stronger contraction condition

$$\mathcal{L}V(x) \leq -\alpha V(x), \quad \alpha > 0,$$

we obtain exponential convergence:

$$\mathbb{E}[V(x(t))] \leq e^{-\alpha t} V(x(0)).$$

6.4 Final structure

The resulting system is:

- a latent Gaussian state $z(t)$ with uncertainty,
- a softmax-mapped probability state $x(t) \in \Delta_{16}$,
- endogenous Fisher-information weights defining the geometry,
- a KL governance potential aligned with Fisher geometry,
- controlled stochastic dynamics on latent space,
- strict generator dissipation ensuring convergence to x^* .

This closes the remaining degrees of freedom: measurement, representation, dynamics, and governance are now jointly specified in a controlled information-geometric stochastic manifold.

7 Final Hybrid Human–AI Organizational Theory

7.1 Canonical Probability Embedding Unit Interval Normalization

Let $x(t) \in [0, 1]^{17}$ satisfy the Fisher-weighted simplex constraint

$$\sum_{i=1}^{17} w_i x_i(t) = 1, \quad w_i > 0, \quad \sum_{i=1}^{17} w_i = 1.$$

Define the canonical probability embedding

$$p_i(x) = w_i x_i, \quad i = 1, \dots, 17.$$

Thus $p(x) \in \Delta^{16}$ (the probability simplex).

7.2 Governance-Optimal Equilibrium

Let $C : \mathcal{X} \rightarrow \mathbb{R}$ be a measurable governance cost functional. Define the governance-optimal state

$$x^* = \arg \min_{x \in \mathcal{X}} \mathbb{E}[C(x)].$$

Assume strict interiority (non-degeneracy):

$$p_i(x^*) \geq \varepsilon > 0, \quad \forall i.$$

Define the relative interior:

$$\mathcal{X}^\circ = \{x \in \mathcal{X} : p_i(x) > 0 \ \forall i\}.$$

7.3 Governance Potential (KL Lyapunov Function)

Define the governance potential

$$V(x) = D_{\text{KL}}(p(x) \| p(x^*)) = \sum_{i=1}^{17} p_i(x) \log \frac{p_i(x)}{p_i(x^*)}.$$

Then

$$V(x) \geq 0, \quad V(x) = 0 \iff x = x^*.$$

Moreover, at equilibrium:

$$\nabla^2 V(x^*) = \mathcal{I}(x^*),$$

where $\mathcal{I}$ is the Fisher information matrix, yielding local information-geometric structure.

7.4 Admissible Controls

Define the admissible control set

$$\mathcal{U}_{\text{adm}} = \{u(t) : \|u(t)\| \leq M, \ u(t) \text{ progressively measurable w.r.t. } \mathcal{F}_t\}.$$

This set is closed, convex, bounded, and non-anticipatory.

7.5 Tangent Stochastic Dynamics

The state evolves via the controlled SDE

$$dx(t) = f(x(t), u(t); \theta) dt + \Sigma(x(t); \theta) dW_t,$$

subject to tangent constraints ensuring invariance of $\mathcal{X}$:

$$\sum_{i=1}^{17} w_i f_i(x, u; \theta) = 0, \quad \sum_{i=1}^{17} w_i \Sigma_{ij}(x; \theta) = 0, \quad \forall j.$$

Assume f, Σ are globally Lipschitz with linear growth.

7.6 Infinitesimal Generator

Let $\mathcal{L}$ denote the generator:

$$\mathcal{L}V(x) = \nabla V(x)^\top f(x, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma(x; \theta) \Sigma(x; \theta)^\top \nabla^2 V(x)).$$

7.7 Strict Governance Dissipation

Assume strict negativity:

$$\mathcal{L}V(x) < 0 \quad \forall x \neq x^*.$$

Stronger contraction condition:

$$\mathcal{L}V(x) \leq -\alpha V(x), \quad \alpha > 0.$$

7.8 Exponential Convergence

Then:

$$\mathbb{E}[V(x(t))] \leq e^{-\alpha t} V(x(0)).$$

Thus the system exhibits exponential convergence in KL divergence.

7.9 Final Theorem (Governance Contraction Theorem)

Assume:

- Fisher-weighted simplex constraint,
- canonical probability embedding $p_i(x) = w_i x_i$,
- strictly interior equilibrium x^* ,
- KL governance potential $V(x)$,
- globally Lipschitz tangent SDE dynamics,
- admissible control set $\mathcal{U}_{\text{adm}}$,
- strict generator dissipation $\mathcal{L}V(x) \leq -\alpha V(x)$,

then for any initial condition $x(0) \in \mathcal{X}^\circ$:

$$x(t) \rightarrow x^* \quad \text{almost surely and in probability,}$$

and

$$\mathbb{E}[D_{\text{KL}}(p(x(t)) \parallel p(x^*))] \leq e^{-\alpha t} D_{\text{KL}}(p(x(0)) \parallel p(x^*)).$$

Hence, the system is:

- measurable,
- constrained on a compact simplex manifold,
- identifiable via trajectory distributions,
- information-geometric via Fisher–KL structure,
- governance-determined via x^* ,
- exponentially convergent,
- and empirically falsifiable.

8 Canonical Embedding, Governance Potential, Generator, Convergence (Latent–Probabilistic Version)

8.1 Governance potential as KL divergence

In the latent–probabilistic formulation, the organizational state is

$$z(t) \sim \mathcal{N}(\mu(t), \Sigma(t)), \quad x(t) = p(t) = \text{softmax}(z(t)) \in \Delta_{16}.$$

Let x^* denote the governance-optimal equilibrium, defined as the minimizer of a governance cost functional:

$$x^* = \arg \min_{x \in \Delta_{16}} \mathbb{E}[C(x)].$$

Assume strict interiority:

$$p_i(x^*) \geq \varepsilon > 0, \quad \forall i,$$

ensuring that KL divergence is finite and differentiable.

8.2 KL governance potential

Define the governance potential

$$V(x) = D_{\text{KL}}(p(x) \parallel p(x^*)) = \sum_{i=1}^{17} p_i(x) \log \frac{p_i(x)}{p_i(x^*)}.$$

This potential satisfies:

$$V(x) \geq 0, \quad V(x) = 0 \iff x = x^*.$$

At equilibrium, Fisher–KL duality holds:

$$\nabla^2 V(x^*) = \mathcal{I}(x^*),$$

where $\mathcal{I}$ is the Fisher information matrix of $p(x)$.

8.3 Generator condition and convergence

The dynamics evolve on latent space:

$$dz(t) = f_z(z(t), u(t); \theta) dt + \Sigma_z(z(t); \theta) dW_t.$$

By Itô's lemma, the generator acting on $V(x)$ is

$$\mathcal{L}V(x) = \nabla_z V(x)^\top f_z(z, u; \theta) + \frac{1}{2} \text{Tr}(\Sigma_z(z; \theta) \Sigma_z(z; \theta)^\top \nabla_z^2 V(x)).$$

We impose the governance stability law:

$$\mathcal{L}V(x) \leq 0 \quad \forall x \in \Delta_{16}, \quad u \in \mathcal{U}_{\text{adm}}.$$

Under strict negativity,

$$\mathcal{L}V(x) < 0 \quad \forall x \neq x^*,$$

the system satisfies

$$x(t) \rightarrow x^* \quad \text{almost surely and in probability.}$$

Under the stronger contraction condition

$$\mathcal{L}V(x) \leq -\alpha V(x), \quad \alpha > 0,$$

we obtain exponential convergence:

$$\mathbb{E}[V(x(t))] \leq e^{-\alpha t} V(x(0)).$$

8.4 Final structure

The resulting system is:

- a latent Gaussian state $z(t)$ with uncertainty,
- a softmax-mapped probability state $x(t) \in \Delta_{16}$,
- endogenous Fisher-information weights defining the geometry,
- a KL governance potential aligned with Fisher geometry,
- controlled stochastic dynamics on latent space,
- strict generator dissipation ensuring convergence to x^* .

This closes the remaining degrees of freedom: measurement, representation, dynamics, and governance are now jointly specified in a controlled information-geometric stochastic manifold.

9 Discussion

The development of this framework demonstrates that hybrid Human–AI organizations can be represented through a unified probabilistic structure without sacrificing interpretability or mathematical rigor. The transition from unit-interval normalization to a latent probabilistic representation was central to this outcome. Although the unit-interval approach offered initial simplicity, it proved insufficient for large, heterogeneous, and data-rich organizational environments. It could not express uncertainty, mixed data types, or information-geometric structure, and it imposed artificial constraints that distorted the underlying organizational dynamics.

The latent probabilistic model resolves these limitations by embedding each organizational dimension into a latent space and mapping the resulting

vector to the probability simplex through a softmax transformation. This approach naturally supports mixed qualitative and quantitative data, enables uncertainty-aware modeling, and provides a canonical geometric structure through the Fisher information metric. The resulting probability simplex is not an imposed constraint but an intrinsic property of the representation.

The integration of RKHS-bounded feature maps ensures that measurement is closed, non-arbitrary, and statistically estimable from the organizational universe. The controlled stochastic differential equation on latent space captures the evolution of hybrid organizational behavior under both human and AI influences. Governance enters the system through a Kullback–Leibler potential, which acts as a Lyapunov function and guarantees strict dissipation toward a governance-optimal equilibrium.

Together, these components form a controlled stochastic manifold with explicit information geometry. The model is expressive enough to capture complex organizational behavior, yet simple enough to be interpretable, testable, and operationally deployable. The framework also provides a foundation for future extensions, including adaptive governance, multi-agent coordination, and real-time organizational diagnostics in highly advance area of Organization Theory and Design, Corporate Governance and Value creation through Strategic Management.

10 Conclusion

This work establishes the first mathematically closed, fully identifiable, and empirically falsifiable theory of hybrid Human–AI organizational design. By reducing organizational complexity to a unified latent probabilistic representation, the framework resolves long-standing challenges in measurement, dynamics, constraints, and governance. The softmax-mapped latent space provides a natural probability simplex equipped with Fisher–KL information geometry, while RKHS-bounded feature maps ensure principled and data-driven measurement. The controlled stochastic dynamics on latent space capture the evolution of organizational behavior, and the KL governance potential guarantees convergence to a governance-optimal equilibrium.

The transition from unit-interval normalization to a latent probabilistic model was essential for achieving generality, interpretability, and mathematical coherence. The resulting structure is compact yet expressive, simple yet rigorous, and capable of supporting diverse organizational configurations. It

offers a robust foundation for analyzing, designing, and governing AI-native organizations in a principled and scientifically testable manner.

Future work may extend this framework to multi-level governance, distributed organizational architectures, and real-time adaptive control. Nonetheless, the core contribution of this research is clear: hybrid Human–AI organizations can be represented through a single, indestructible mathematical structure that unifies information geometry, stochastic dynamics, and governance into a coherent whole.

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Appendix A. Classical Organizational Design Dimensions

This appendix provides mathematically rigorous definitions of the eleven classical organizational design dimensions commonly used in organizational theory. Each dimension is expressed as a measurable coordinate of the organizational state vector

$$x(t) = (F(t), S_p(t), H(t), C(t), P(t), R(t), G_s(t), E(t), Z(t), U(t), T(t)) \in \mathbb{R}^{11},$$

where each coordinate is defined as a conditional expectation of a feature map applied to organizational data available at time t . These definitions generalize across organizational types and support integration into latent probabilistic or information-geometric models.

A.1 Formalization $F(t)$

Formalization quantifies the extent to which organizational behaviour is governed by explicit written rules, procedures, and documentation.

Let $\mathcal{D}(t)$ denote the corpus of valid internal documents at time t . Let $\varphi_F : \mathcal{D} \rightarrow \mathbb{R}$ be a feature map extracting a proxy for rule density (e.g., coverage, length, or structural complexity). Then

$$F(t) = \mathbb{E}[\varphi_F(\mathcal{D}(t)) \mid \mathcal{F}_t].$$

Higher values of $F(t)$ indicate greater reliance on formal written guidance.

A.2 Specialization $S_p(t)$

Specialization measures the degree to which work is subdivided into distinct roles.

Let $\mathcal{J}(t)$ be the set of jobs at time t , and let $\tau(j)$ denote the task breadth or diversity of job j . For a decreasing function g ,

$$S_p(t) = \mathbb{E}[g(\tau(j)) \mid j \in \mathcal{J}(t), \mathcal{F}_t].$$

Higher $S_p(t)$ corresponds to narrower roles and finer division of labour.

A.3 Hierarchy of Authority $H(t)$

Hierarchy captures vertical layering and span of control.

Let $L(t)$ be the number of reporting levels and $s_k(t)$ the span of control at level k . For an aggregation function h ,

$$H(t) = h\left(L(t), \{s_k(t)\}_{k=1}^{L(t)}\right),$$

with $H(t)$ increasing in $L(t)$ and decreasing in average span.

A.4 Centralization $C(t)$

Centralization reflects where decision authority is located in the hierarchy.

Let $\mathcal{D}_{\text{dec}}$ be a set of decision types. For each decision $d \in \mathcal{D}_{\text{dec}}$, let $\ell(d, t)$ denote the organizational level responsible for that decision. Then

$$C(t) = \mathbb{E}[\ell(d, t) \mid d \in \mathcal{D}_{\text{dec}}, \mathcal{F}_t],$$

normalized so that higher $C(t)$ indicates more top-level concentration.

A.5 Professionalism $P(t)$

Professionalism measures the formal education or training required for roles.

Let $Y_i(t)$ denote the education or training index of employee i at time t . Then

$$P(t) = \mathbb{E}[Y_i(t) \mid i \in \mathcal{E}(t), \mathcal{F}_t],$$

where $\mathcal{E}(t)$ is the employee set. Higher $P(t)$ indicates a more professionally trained workforce.

A.6 Personnel Ratios $R(t)$

Personnel ratios describe how employees are distributed across functional categories.

Partition employees into categories $\mathcal{C}$ (e.g., administrative, professional, direct, indirect). For category $c \in \mathcal{C}$,

$$r_c(t) = \frac{N_c(t)}{N_{\text{total}}(t)},$$

and the personnel ratio vector is

$$R(t) = (r_c(t))_{c \in \mathcal{C}}.$$

Appendix B. Contextual Dimensions

B.1 Goals and Strategy $G_s(t)$

Goals and strategy represent the intended direction and competitive posture.

Let $C(x)$ be a governance or performance cost functional. Define the strategic target

$$x^* = \arg \min_x \mathbb{E}[C(x)].$$

Then alignment is measured by

$$G_s(t) = -D(x(t) \parallel x^*),$$

where D is a divergence. Higher $G_s(t)$ indicates stronger strategic alignment.

B.2 Environment $E(t)$

Environment captures external conditions influencing the organization.

Let $\mathcal{E}_{\text{ext}}(t)$ denote external signals (industry, regulation, customers, competitors). Let φ_E summarize environmental complexity or volatility. Then

$$E(t) = \mathbb{E}[\varphi_E(\mathcal{E}_{\text{ext}}(t)) \mid \mathcal{F}_t].$$

B.3 Size $Z(t)$

Size reflects the magnitude of the organization as a social system.

Let $N_{\text{total}}(t)$ be total headcount. Then

$$Z(t) = \log(1 + N_{\text{total}}(t)),$$

or another monotone transformation.

B.4 Culture $U(t)$

Culture represents shared values, norms, and behavioural patterns.

Let $\mathcal{C}_t$ denote cultural artifacts (language, symbols, behaviours). Let $\varphi_U(\mathcal{C}_t)$ be an embedding into a latent representation, and let $u(t)$ be a scalar index extracted from it. Then

$$U(t) = \mathbb{E}[u(t) \mid \mathcal{F}_t].$$

B.5 Technology $T(t)$

Technology describes the tools and processes used to transform inputs into outputs.

Let $\mathcal{K}(t)$ denote the set of production and information technologies. Let $\varphi_T(\mathcal{K}(t))$ measure technological sophistication or integration. Then

$$T(t) = \mathbb{E}[\varphi_T(\mathcal{K}(t)) \mid \mathcal{F}_t].$$

Appendix C. AI-Native Organizational Dimensions

This appendix defines six AI-native organizational dimensions that complement the eleven classical dimensions presented in Appendices A and B. These dimensions capture the structural, cognitive, and governance-related effects of AI systems within hybrid Human–AI organizations. Each dimension is expressed as a measurable coordinate of the AI state vector

$$x_{\text{AI}}(t) = (A_p(t), A_u(t), D_s(t), K_c(t), L_r(t), G_c(t)) \in \mathbb{R}^6,$$

which integrates with the classical state vector to form the full 17-dimensional hybrid organizational state space.

C.1 AI Penetration $A_p(t)$

AI penetration measures the extent to which AI systems participate in core organizational processes.

Let $\mathcal{W}(t)$ denote the set of work processes at time t , and let $\alpha(w, t) \in [0, 1]$ represent the proportion of each process w executed or supported by AI systems. Then

$$A_p(t) = \mathbb{E}[\alpha(w, t) \mid w \in \mathcal{W}(t), \mathcal{F}_t].$$

Higher $A_p(t)$ indicates deeper integration of AI into operational workflows.

C.2 AI Autonomy $A_u(t)$

AI autonomy captures the degree to which AI systems act without direct human intervention.

Let $\mathcal{A}(t)$ be the set of AI agents active at time t , and let $\beta(a, t) \in [0, 1]$ denote the autonomy index of agent a , reflecting its ability to make decisions, adapt, or self-correct. Then

$$A_u(t) = \mathbb{E}[\beta(a, t) \mid a \in \mathcal{A}(t), \mathcal{F}_t].$$

Higher $A_u(t)$ corresponds to more independent and self-directed AI behavior.

C.3 Decision Share $D_s(t)$

Decision share quantifies the proportion of organizational decisions made or co-made by AI systems.

Let $\mathcal{D}_{\text{org}}(t)$ denote the set of organizational decisions at time t , and let $\delta(d, t) \in [0, 1]$ represent the AI contribution to decision d . Then

$$D_s(t) = \mathbb{E}[\delta(d, t) \mid d \in \mathcal{D}_{\text{org}}(t), \mathcal{F}_t].$$

Higher $D_s(t)$ indicates greater AI involvement in decision-making processes.

C.4 Coordination Density $K_c(t)$

Coordination density measures the intensity of interactions between human and AI agents.

Let $\mathcal{I}(t)$ denote the set of Human–AI interaction events at time t , and let $\kappa(i, t)$ quantify the informational or operational load of interaction i . Then

$$K_c(t) = \mathbb{E}[\kappa(i, t) \mid i \in \mathcal{I}(t), \mathcal{F}_t].$$

Higher $K_c(t)$ reflects denser or more complex Human–AI coordination.

C.5 Cognitive Load Redistribution $L_r(t)$

Cognitive load redistribution captures how AI systems shift cognitive effort between humans and machines.

Let $\lambda_h(t)$ and $\lambda_a(t)$ denote the human and AI cognitive load indices, respectively. Define the redistribution index as

$$L_r(t) = \lambda_a(t) - \lambda_h(t),$$

normalized so that positive values indicate net transfer of cognitive burden from humans to AI systems.

C.6 Governance Constraint $G_c(t)$

Governance constraint measures the strength of oversight, alignment, and compliance mechanisms applied to AI systems.

Let $\pi(t)$ denote the AI policy distribution and π^* the governance target distribution. Define

$$G_c(t) = D_{\text{KL}}(\pi(t) \parallel \pi^*),$$

where D_{KL} is the Kullback–Leibler divergence. Lower $G_c(t)$ indicates stronger alignment with governance requirements.